7	**(a)**	**(i)**	nucleus contains 92 protons	B1	
			nucleus contains 143 neutrons (missing 'nucleus' 1/2)	B1	
			outside / around nucleus 92 electrons	(B1)	
			most of atom is empty space / mass concentrated in nucleus	(B1)	
			total charge is zero	(B1)	
			diameter of atom ~ 10^{-10} m or size of nucleus ~ 10^{-15} m	(B1)	
			any two of (B1) marks		[4]
		(ii)	nucleus has same number / 92 protons	B1	
			nuclei have 143 and 146 neutrons (missing 'nucleus' 1/2)	B1	[2]
	(b)	**(i)**	$Y = 35$	A1	
			$Z = 85$	A1	[2]
		(ii)	mass-energy is conserved in the reaction	B1	
			mass on rhs of reaction is less so energy is released explained in terms of $E = mc^2$	B1	[2]